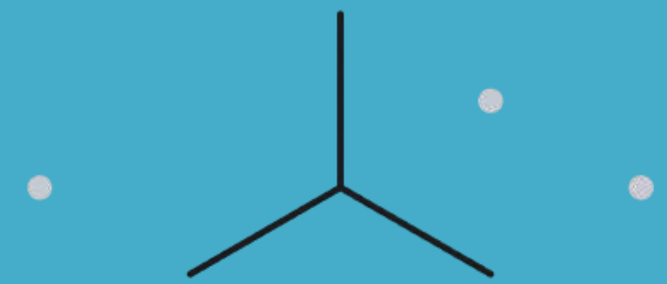
CARD B01 - NAME → STRUCTURE

FOUNDATION LEVEL: GIVEN NAME: 2-METHYLPROPANE

1. NAME DECODED

- Parent: propane, 3 carbon atoms, open chain.
- Substituent: methyl (CH_3) at carbon 2.
- expected formula:Carbon 2 has one hydrogen because it is connected to three carbons. The other three carbons have three hydrogens each. Therefore, the formula is C_4H_{10}.

2. BOND LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

- Formula counted from structure: C_4H_{10}
- General formula result: C_4H_{10}
- Verdict: Match.

4. BACK-CHECK

- Name returned by The Alkane: 2-methylpropane
- Verdict: Match

✓ parent length matches ✓ substituent position match ✓ formula agrees ✓ back-check recorded

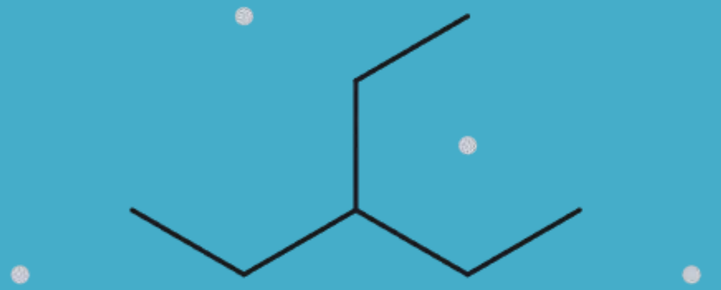
CARD B02 - NAME → STRUCTURE

FOUNDATION LEVEL: GIVEN NAME: 3-ETHYLPENTANE

1. NAME DECODED

- Parent: pentane, 5 carbon atoms, open chain.
- Substituent: ethyl (CH_2CH_3) at carbon 3.
- expected formula: Carbon 3 has one hydrogen, the two end carbons have three each, and the remaining four carbons have two or three as needed. Total: 7 carbons and 16 hydrogens. Therefore, C_7H_{16}.

2. BOND LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

- Formula counted from structure: C_7H_{16}
- General formula result: C_7H_{16}
- Verdict: Match.

4. BACK-CHECK

- Name returned by The Alkane: 3-ethylpentane
- Verdict: Match

✓ parent length matches ✓ substituent position match ✓ formula agrees ✓ back-check recorded

CARD BO3 - NAME → STRUCTURE

FOUNDATION LEVEL: GIVEN NAME: 2,2-DIMETHYLBUTANE

1. NAME DECODED

- Parent: butane, 4 carbon atoms, open chain.
- Substituents: two methyl groups at carbon 2.
- expected formula: Carbon 2 has no hydrogens because it is connected to four carbons. The three methyl groups have three hydrogens each, while the remaining carbons complete the total. Therefore, C_6H_{14}.

2. BOND LINE STRUCTURE (THE ALKANE)

3. FORMULA CHECK

- Formula counted from structure: C_6H_{14}
- General formula result: C_6H_{14}
- Verdict: Match.

4. BACK-CHECK

- Name returned by The Alkane: 2,2-dimethylbutane
- Verdict: Match

✓ parent length matches ✓ substituent position match ✓ formula agrees ✓ back-check recorded

CARD B04 - NAME → STRUCTURE

DEVELOPING LEVEL: GIVEN NAME: 3-ETHYL-2-METHYLPENTANE

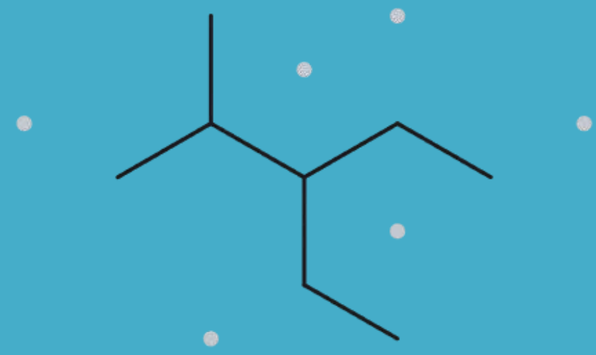
1. NAME DECODED

- Parent: pentane, 5 carbon atoms, open chain.
- Substituents:

-Ethyl (CH_2CH_3) at carbon 3.
-Methyl (CH_3) at carbon 2.

- expected formula: Carbons 2 and 3 each have one hydrogen because they are connected to three carbons. The remaining carbons have two or three hydrogens. Therefore, C_8H_{18}.

2. BOND LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

- Formula counted from structure: C_8H_{18}.
- General formula result: C_8H_{18}.
- Verdict: Match.

4. BACK-CHECK

- Name returned by The Alkane: 3-ethyl-2-methylpentane
- Verdict: Match

✓ parent length matches ✓ substituent position match ✓ formula agrees ✓ back-check recorded

CARD B05 - NAME → STRUCTURE

DEVELOPING LEVEL: GIVEN NAME: 1-CHLORO-2-METHYLPROPANE

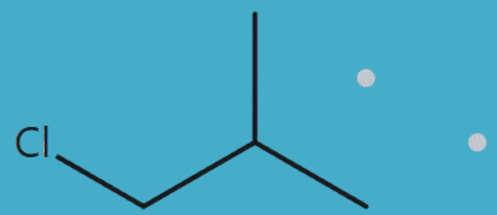
1. NAME DECODED

- Parent: propane, 3 carbon atoms, open chain.
- Substituents:

-Chloro (Cl) at carbon 1.
-Methyl (CH_3) at carbon 2.

- Expected formula: Carbon 1 has two hydrogens, carbon 2 has one, and the two methyl carbons have three each. Including chlorine, the formula is C_4H_9Cl.

2. BOND LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

- Formula counted from structure: C_4H_9Cl.
- General formula result after halogen substitution: C_4H_9Cl.
- Verdict: Match.

4. BACK-CHECK

- Name returned by The Alkane: 1-chloro-2-methylpropane
- Verdict: Match

✓ parent length matches ✓ substituent position match ✓ formula agrees ✓ back-check recorded

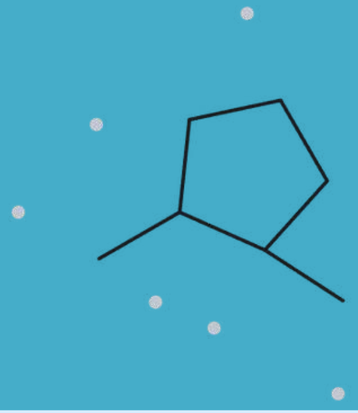
CARD BO6 - NAME → STRUCTURE

DEVELOPING LEVEL: GIVEN NAME: 1,2-DIMETHYLCYCLOPENTANE

1. NAME DECODED

- Parent: cyclopentane, 5 carbon atoms, one ring.
- Substituents: two methyl groups at carbons 1 and 2.
- Expected formula: Ring carbons 1 and 2 each have one hydrogen. The other three ring carbons have two each, and the methyl groups have three each. Therefore, C_7H_{14}.

2. BOND LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

- Formula counted from structure: C_7H_{14}.
- General formula result: C_7H_{14}.
- Verdict: Match.

4. BACK-CHECK

- Name returned by The Alkane: 1,2-dimethylcyclopentane
- Verdict: Match

✓ parent length matches ✓ substituent position match ✓ formula agrees ✓ back-check recorded

CARD B07 - NAME → STRUCTURE

DEVELOPING LEVEL: GIVEN NAME: 4-ETHYL-3,3-DIMETHYLHEPTANE

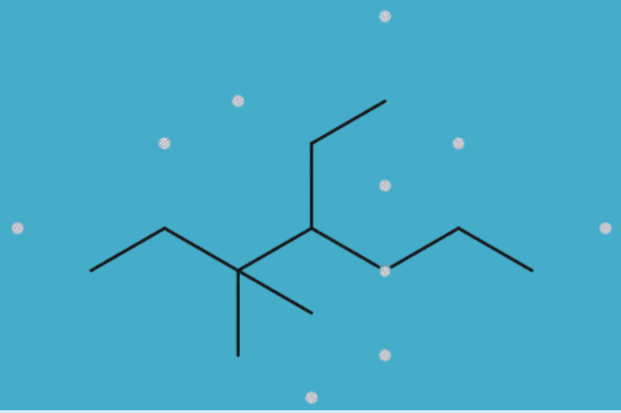
1. NAME DECODED

- Parent: heptane, 7 carbon atoms, open chain.
- Substituents:

-Ethyl (CH_2CH_3) at carbon 4.
-Two methyl groups at carbon 3.

- Expected formula:Carbon 3 has no hydrogens, while carbon 4 has one. The remaining carbons have two or three hydrogens. Therefore, $C_{11}H_{24}$.

2. BOND LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

- Formula counted from structure: $C_{11}H_{24}$.
- General formula result: $C_{11}H_{24}$.
- Verdict: Match.

4. BACK-CHECK

- Name returned by The Alkane: 4-ethyl-3,3-dimethylheptane
- Verdict: Match

✓ parent length matches ✓ substituent position match ✓ formula agrees ✓ back-check recorded

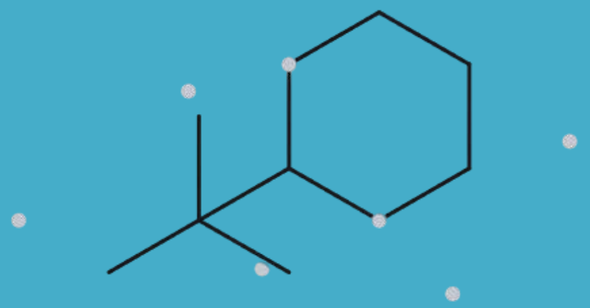
CARD B08 - NAME → STRUCTURE

CHALLENGE LEVEL: GIVEN NAME: TERT-BUTYLCYCLOHEXANE

1. NAME DECODED

- Parent: cyclohexane, 6 carbon atoms, one ring.
- Substituent: tert-butyl, attached to the ring.
- The tert-butyl group contains one central carbon bonded to three methyl groups.
- A locant is not necessary because there is only one substituent on the ring.
- Expected formula:The attached ring carbon has one hydrogen, while the other five ring carbons have two each. The central tert-butyl carbon has none, and its three methyl groups have three each. Therefore, $C_{10}H_{20}$.

2. BOND LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

- Formula counted from structure: $C_{10}H_{20}$.
- General formula result: $C_{10}H_{20}$.
- Verdict: Match.

4. BACK-CHECK

- Name returned by The Alkane:(1,1-dimethylethyl)cyclohexane
- Verdict: does not match
- reason: they dont match in name only but they represent the same compound

✓ parent length matches ✓ substituent position match ✓ formula agrees ✓ back-check recorded

CARD B09 - NAME → STRUCTURE

CHALLENGE LEVEL: GIVEN NAME: 3-ETHYL-1,1-DIMETHYLCYCLOHEXANE

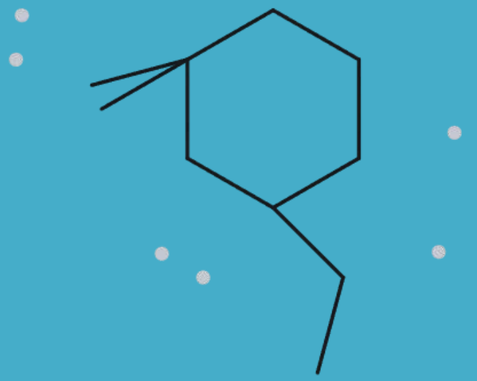
1. NAME DECODED

- Parent: cyclohexane, 6 carbon atoms, one ring.
- Substituents:

-Two methyl groups at carbon 1.
-One ethyl group at carbon 3.

- Expected formula:Carbon 1 has no hydrogens, and carbon 3 has one. The other ring carbons have two each, while the branches complete the remaining hydrogens. Therefore, $C_{10}H_{20}$.

2. BOND LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

- Formula counted from structure: $C_{10}H_{20}$.
- General formula result: $C_{10}H_{20}$.
- Verdict: Match.

4. BACK-CHECK

- Name returned by The Alkane: 3-ethyl-1,1-dimethylcyclohexane
- Verdict: does not match

✓ parent length matches ✓ substituent position match ✓ formula agrees ✓ back-check recorded

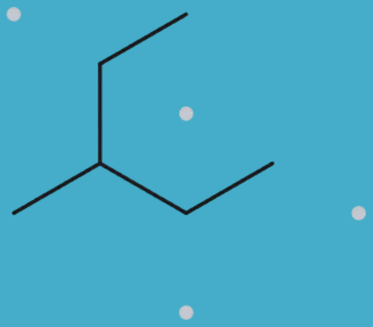
CARD B1O - NAME → STRUCTURE

CHALLENGE LEVEL: GIVEN NAME: 2-ETHYLBUTANE

1. NAME DECODED

- Given parent: butane, 4 carbon atoms, open chain.
- Given substituent: ethyl (CH_2CH_3) at carbon 2.
- Expected formula: The central carbon has one hydrogen, the two end carbons have three each, and the remaining carbons have two or three. Therefore, C_6H_{14}.

2. BOND LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

- Formula counted from structure: C_6H_{14}.
- General formula result: C_6H_{14}.
- Verdict: Match.

4. BACK-CHECK

- Name returned by The Alkane: 3-methylpentane
- Verdict: does not match
- reason: It broke the longest chain rule

✓ parent length matches ✓ substituent position match ✓ formula agrees ✓ back-check recorded